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Original Article

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Genomic contingence beneath ecological convergence: the tempo and mode of gene loss in parasitic bilaterians

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Abstract

Parasitism has independently evolved hundreds of times among metazoans. Nonetheless, parasites have explored only a limited range of ecologies, and they display frequent convergence in morphological, behavioral, and life-history traits. Although gene loss in particular parasitic species has been documented, it is not known if gene loss converges along the same lines as these other traits. To test for convergent gene loss, we characterized the housekeeping, regulatory, and DNA-repair complements of 48 bilaterian species, including 20 parasites belonging to 6 different bilaterian phyla. We found that different parasitic strategies do not display characteristic tempos or modes of gene loss. Further, the accelerated rates of gene loss seen in some parasites were almost always shared with their free-living relatives, indicating that the increased rate of loss preceded the rise of parasitism. Therefore, the convergent ecological strategies and adaptations that have arisen in distantly related parasitic lineages overlay contingent gene losses, which largely reflect their phylogenetic history. These results have important implications for how ecologists and evolutionary biologists should model the acquisition of parasitism, especially regarding the long-held assumption that reversion from a parasitic to a free-living state is impossible.

Keywords: parasitism, convergence, contingence, macroevolution, comparative genomics

Introduction

Though more than 200 animal lineages have independently evolved a parasitic existence (Dobson et al., 2008; Weinstein & Kuris, 2016), the range of parasitic lifestyles is remarkably restricted, with distantly related species consistently converging on a limited suite of ecologies (Poulin, 2011; Poulin & Randhawa, 2015). Epidemiologists first systematized this convergence by separating macroparasites from microparasites according to whether the virulence of an infection depended on its initial intensity (Anderson & May, 1979; May & Anderson, 1979). Later, Kuris and Lafferty (2000) and Lafferty and Kuris (2002) proposed eight parasitic strategies based on intensity dependence, the number of hosts parasitized, whether at least one host needed to die, and the severity of host fitness reduction. More recently, Poulin (2011) argued that intensity dependence is a phylogenetically fixed characteristic that cannot define a convergent ecological strategy. The removal of this contrast left six strategies: parasitoidism, parasitic castration, direct transmission, trophic transmission, vector transmission, and micropredation (Figure 1) (Poulin, 2011; Poulin & Randhawa, 2015).

Each of Poulin's strategies is associated with a characteristic suite of convergent adaptations (Poulin, 2011; Poulin & Randhawa, 2015), such as the suppression of host sexual dimorphism by castrators (Baudoin, 1975; Hall et al., 2007; Lafferty & Kuris, 2009; Reinhard, 1956), the manipulation of host behavior by trophically transmitted parasites (Bethel & Holmes, 1973; Combes, 1991; Kuris, 2003; Lafferty,

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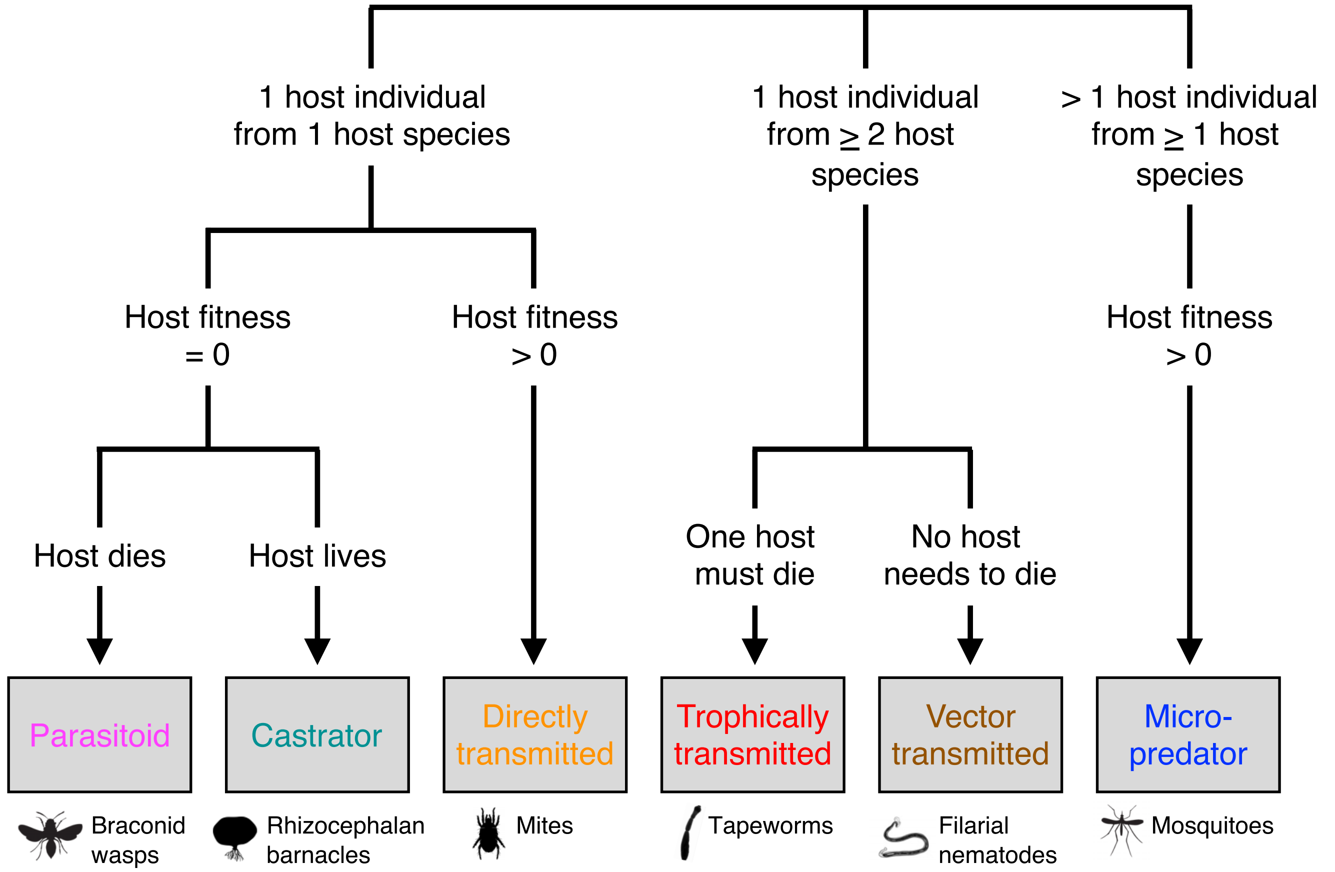

Figure 1. The six parasitic strategies recognized by Poulin (2011) and Poulin and Randhawa (2015), along with taxonomic examples for each of the six strategies. Parasitoids infect one host during their lifecycle; the host's fitness is reduced to 0 because it is killed as part of the parasite's development. Parasitic castrators also infect just one host and reduce its fitness to 0, but this reduction occurs because the castrator either directly or indirectly destroys the host's gonads. Directly transmitted parasites infect just one host but do not individually reduce the fitness of that host to 0. Trophically transmitted and vector-transmitted parasites sequentially infect at least two hosts (i.e., a single lifecycle stage typically occurs entirely in only one host) from different species. Trophic transmission occurs when an uninfected definitive host consumes an infected intermediate host; vector transmission occurs when an infected intermediate host (typically a micropredatory insect) bites an uninfected definitive host. Micropredators take small meals from many different hosts, often from several different species, over the course of their lifecycles. Figure modified from *Advances in Parasitology*,74, Poulin, "The Many Roads to Parasitism: A Tale of Convergence," pp. 1-40, 2011, with permission from Elsevier (Poulin, 2011).

1999; Poulin, 1994), and the modification of heat-shock and carboxylesterase genes in blood-feeding micropredators (Benoit et al., 2011; Freitas & Nery, 2020; Paim et al., 2016). However, few studies have tested for convergent gene losses within each strategy, or for differences between loss in parasites and their close free-living relatives, despite the historical assumption that parasites are simplified. Indeed, though a substantial literature documents gene loss in individual parasitic species or clades (e.g., Cunha et al., 2023; Fromm et al., 2013; Herlyn et al., 2025; International Helminth Genomes Consortium, 2019; Kirkness et al., 2010; Olson et al., 2012; Tsai et al., 2013), we are not aware of any work that rigorously considered ecological convergence as a variable shaping the tempo or mode of gene loss among a sample of distantly related parasites.

Peterson et al. (2026) incidentally catalogued gene losses across a broad sample of largely free-living bilaterians while examining the link between the acquisition of novel regulatory genes and the rise of novel cell types. This study did not extensively consider patterns of loss, even among free-living species, and included just eight parasites representing only four strategies, so it could not draw broadly generalizable conclusions about patterns of loss in parasites. However, it did find that all sampled bilaterians, including the parasites, displayed the same relative degree of regulatory and housekeeping gene loss, suggesting that loss in parasites may be more similar to other organisms than is widely assumed. This unexpected result notwithstanding, the tempo and many dimensions of the mode (e.g., patterns of gene identity) of loss in parasites were not addressed by that study. Thus, more extensive sampling and more comprehensive analyses are needed to determine whether parasites display patterns of gene loss which distinguish them from their free-living relatives or uniquely characterize particular strategies. In this context, we present the first broad-based assessment of the tempo and mode of gene loss in parasites that compares these species with their free-living relatives and explicitly examines convergence within strategies.

To test for convergence in the tempo and mode of gene loss associated with Poulin's strategies, we annotated the housekeeping, regulatory, and DNA-repair protein repertoires of an additional 12 parasitic and 10 free-living species, including four representatives of the two previously unsampled strategies. Using this total sample of more than 32,000 sequences from 48 species across 12 phyla, we confirmed that all bilaterians, regardless of ecology, lose housekeeping and

Table 1. Ecological information for each of the parasites examined in this study, with newly sampled species bolded.

Species	Common name	Strategy	Parasitic life stage	Definitive host
Raillietiella orientalis	**Tongue worm**	**Trophically transmitted**	**Lifelong**	**Vertebrata**
Sacculina carcini	**Crab-hacker barnacle**	**Castrator**	**Adult**	**Arthropoda**
Lepeophtheirus salmonis	**Salmon louse**	**Directly transmitted**	**Adult**	**Vertebrata**
Tetranychus urticae	Two-spotted spider mite	Directly transmitted	Lifelong	Plantae
Diachasmimorpha longicaudata	Braconid wasp	Parasitoid	Larval	Arthropoda
Anopheles gambiae	Malaria mosquito	Micropredator	Adult (f), Free (m)	Vertebrata
Glossina pallidipes	**Tsetse fly**	**Micropredator**	**Adult**	**Vertebrata**
Aphis gossypii	**Cotton aphid**	**Micropredator**	**Lifelong**	**Plantae**
Ixodes scapularis	Deer tick	Micropredator	Lifelong	Vertebrata
Xenos peckii	**Twisted-wing insect**	**Castrator**	**Lifelong (f), Larval (m)**	**Arthropoda**
Brugia malayi	**Filarial nematode**	**Vector-transmitted**	**Lifelong**	**Vertebrata**
Ascaris suum	**Large roundworm of pigs**	**Directly transmitted**	**Lifelong**	**Vertebrata**
Romanomermis culicivorax	**Mermithid nematode**	**Parasitoid**	**Larval**	**Arthropoda**
Dracunculus medinensis	**Guinea worm**	**Trophically transmitted**	**Lifelong**	**Vertebrata**
Bursaphelenchus xylophilus	**Pine wood nematode**	**Vector-transmitted**	**Lifelong**	**Plantae**
Gordionus sp.	Horsehair worm	Parasitoid	Larval	Arthropoda
Potamilus streckersoni	**Brazos heelsplitter**	**Directly transmitted**	**Larval**	**Vertebrata**
Echinococcus granulosus	Dog tapeworm	Trophically transmitted	Lifelong	Vertebrata
Schistosoma mansoni	Blood fluke	Directly transmitted	Lifelong	Vertebrata
Pomphorhynchus laevis	Acanthocephalan rotifer	Trophically transmitted	Lifelong	Vertebrata

Note. On the basis of the then-consensus sister relationship between the trematodes and cestodes within the neodermatan platyhelminths, Poulin (2011) argued that the complicated lifecycle of *Schistosoma mansoni* represented a modified form of trophic transmission. However, new phylogenetic hypotheses for the neodermatan platyhelminths (Brabec et al., 2023), and a review of trematode internal relationships and ecology (Loker, 2025) make this interpretation unlikely. We consider *S. mansoni* a serially directly transmitted parasite because transmission between its two hosts does not require any direct interaction between the hosts. In subsequent analyses, species whose parasitic lifecycle stage depends on an individual's sex are coded according to the parasitic stage of the female.

regulatory genes in the same relative proportion. We then placed losses in parasites in a phylogenetic context with the gene complements of their free-living relatives (per Jackson, 2015) and showed for the first time that no parasitic strategies, nor parasites as a whole, are uniquely characterized by enriched losses in particular biological functions, by the loss or retention of a subset of the ancestral bilaterian genome, nor by the loss of any individual genes. Finally, we newly demonstrate that the different parasitic strategies do not have characteristic rates of loss; indeed, almost all parasites with accelerated loss have free-living relatives with similarly accelerated loss. These results have important implications for the evolution of parasitism, especially the application of Dollo's law of irreversible evolution.

Materials and methods

Taxon and gene selection

Taxa were selected according to taxonomic and ecological breadth, plus data availability and quality. When sufficient data existed for several closely related species, we selected the species with the highest BUSCO genome completeness score. We removed the six vertebrates from the original dataset because they are outliers in most aspects of genome evolution (Peterson et al., 2026) and are not closely related to the sampled parasites. Supplementary File 1, tab 1, contains all data accession information for all newly analyzed species.

We processed data for 12 parasites belonging to three bilaterian phyla (Table 1). We sampled the mermithid *Romanomermis culicivorax*, the Guinea worm *Dracunculus medinensis*, the pig roundworm *Ascaris suum*, the filarial worm *Brugia malayii*, and the pine wilt nematode *Bursaphelenchus xylophilus* from the nematodes; the tongue worm *Raillietiella orientalis*, the crab-hacker barnacle *Sacculina carcini*, the salmon louse *Lepeophtheirus salmonis*, the cotton aphid *Aphis gosypii*, the strepsipteran *Xenos peckii*, and the tsetse fly *Glossina pallidipes* from the arthropods; and the Brazos heelsplitter mussel *Potamilus streckersoni* from the molluscs. We also sampled nine free-living relatives of these parasites: *Caenorhabditis elegans*, the marine nematode *Enoplolaimus lenunculus*, and the mycophagous nematode *Aphelenchus avenae* for the nematodes; the ostracod *Darwinula stevensoni* for the tongue worm; the blue crab *Portunus trituberculatus* and the gooseneck barnacle *Pollicipes pollicipes* for the crab-hacker barnacle; the copepod *Tigriopus californicus* for the salmon louse; the water strider *Gerris buenoi* for the aphid; and the red flour beetle *Tribolium castaneum* for the strepsipteran. Finally, we sampled the tardigrade *Paramacrobiotus metropolitanus* because its clade contains few parasites (Nelson et al., 2018) but may have lost many genes (Guijarro-Clarke et al., 2020).

We annotated the BUSCO, regulatory, and DNA repair genes of our species because these genes are closely involved in the phenomena of interest. Regulatory gene products control the developmental process, and DNA repair gene products influence the rate at which genomic novelties arise, so convergent losses of these genes may underly convergent patterns of phenotypic and genomic change. Conversely, convergent ecological circumstances may create parallel opportunities for the loss of the housekeeping and metabolic functions performed by the products of BUSCO genes.

Small RNA dataset preparation

Small RNAs (sRNAs) for the tongue worm *Raillietiella orientalis* were obtained from whole-body samples of two adults (host: *Nerodia fasciata*), a single secondary larva (host: *Anolis sagrei*), and five primary larvae (host: *Blaberus discoidalis*). Wild specimens of the gooseneck barnacle were collected near Aveiro, Portugal. sRNAs for these species were extracted using the Quick-RNA Micro-Prep Kit (Zymo Research) according to the manufacturer's procedure;

libraries were prepared for Illumina NextSeq 500 sequencing using the Nextflex Small RNA Sequencing kit (Revvity).

Wild specimens of the ostracod *Vargula norvegica* were trapped at a depth of 700 m in the Korsfjorden, Norway, and wild specimens of the crab-hacker barnacle (host: *Carcinus maenas*) were collected from the Limfjorden, Denmark. Libraries were isolated from these two species using the NEBNext Small RNA Prep Set (New England Biolabs) for Illumina sequencing. The sRNA dataset for the strepsipteran *Stylops ovinae* was isolated from whole-body specimens of free-living adult males using the small-RNA PAGE Recovery Kit (Zymo Research) and prepared for Illumina HiSeq 2000 sequencing using the TruSeq Small-RNA Library Prep Kit (Illumina).

The tardigrade dataset was retrieved from Campbell et al. (2011); all other species' sRNA libraries were downloaded from the NCBI. No sRNA libraries existed for the ostracod *Darwinula stevensoni*, the marine nematode *Enoplolaimus lenunculus*, or the strepsipteran *Xenos peckii*, so libraries from the related species *Vargula norvegica*, *Enoplus brevis*, and *Stylops ovinae*, respectively, were used. To maximize annotation fidelity, the *S. ovinae* sRNA dataset was analyzed using the genome of *Stylops atterimus*.

microRNA annotation

For the tsetse fly, the red flour beetle, the pig roundworm, and *Caenorhabditis elegans*, microRNA (miRNA) data were taken from MirGeneDB 3.0 (Clarke et al., 2025); for all other taxa, we used the miRNA annotation pipeline established by Ambros et al. (2003) and elaborated by Fromm et al. (2015). Briefly, MirMachine (Umu et al., 2023) predicted the probable members of all known miRNA families present in a given genome. Then, mirTrace (Kang et al., 2018) removed sequences with low PHRED quality scores, low complexity, or a length of fewer than 18 nucleotides from its sRNA dataset. Sequences longer than 28 nucleotides were removed separately. Finally, MirMiner (Wheeler et al., 2009) used the filtered sRNA dataset, genome, and MirMachine predictions to identify probable conserved and novel miRNAs, which were then hand-curated. Paralogous members of known miRNA families were identified using synteny and gene trees created in Unipro UGENE v.52.0 (Golosova et al., 2014; Okonechnikov et al., 2012; Rose et al., 2019) and MacVector v.18.25 (macvector.com/getting-started/macvector-major-release-details-and-history) using the Muscle alignment and neighbor-joining algorithms with default settings.

To confirm losses, the NCBI blastn function (Chen et al., 2015) was used to query each genome for the precursor and mature sequences of all putatively absent ancestral miRNA genes. Then, the relevant sRNA libraries were queried for the seven-nucleotide seed sequence of all mature miRNA sequences not recovered by the blastn query. For a parasitic species, mature sequences identified by the seed search (false-negative miRNAs) were only accepted if they were more similar to the corresponding sequences of its close relatives than those of its host. As a result, losses in species which parasitize their close relatives may be overestimated. No sRNA data were available for the nematodes *Aphelenchus avenae* and *Dracunculus medinensis* or the water strider *Gerris buenoi*, so annotations for these species reflect MirMachine predictions alone. To identify false negatives in *G. buenoi*, all NCBI heteropteran genomes were queried for putative losses using blastn. Similarly, all available podocopan ostracod genomes were used to confirm false negatives in the podocopan ostracod *Darwinula stevensoni* because "Ostracoda" may be polyphyletic (Bernot et al., 2023; Yu et al., 2024). Thus, sequences present in the myodocopan *Vargula norvegica* sRNA dataset but absent from the *Darwinula stevensoni* genome may represent independent losses. Supplementary File 1, tab 2, summarizes the presence–absence data for the 921 miRNA families considered here.

Protein annotation

The transcription factor (TF), RNA-binding protein (RBP), and DNA-repair protein (DNARP) complements of each species were identified using a reciprocal blastp strategy. Briefly, a master file containing sequences of known identity from *Haliotis rufescens* and *Saccoglossus kowalevski* was used in an NCBI blastp search of the predicted proteins for each species. Sequences in the uncharacterized species with significant alignments to the query were then blasted against the full annotated genomes of *Daphnia pulex*, *H. rufesences*, and *S. kowalevski* to confirm their identity. For lineage-specific genes, the query for the initial search and the target sequences for the reciprocal search were selected from a species known to possess that gene. Trees built in UGENE and MacVector were used to clarify uncertain identities. Supplementary File 1, tabs 3–5, summarizes data for each class of protein. These procedures were also used to recover the sequences of 19 proteins used in our molecular clock analysis (see below).

The crab-hacker barnacle, the strepsipteran, the water strider, and the nematodes *Enoplolaimus lenunculus* and *Romanomermis culicivorax* do not have NCBI protein predictions, so published genome annotations (from Martin et al., 2022, Castaño et al., 2024, Armisén et al., 2018, Lee et al., 2023, and Schiffer et al., 2013, respectively) were queried with BLAST+ v.2.17.0 (Camacho et al., 2008) in the initial search, and MetaEuk (Levy Karin et al., 2020) was used to confirm losses. No mRNA data existed for the tongue worm, so all proteins for this species were predicted using computational methods alone. To minimize any resulting overestimation of loss, the protein complement of this species was curated using a BLAST+ search of the combined outputs of three independent BUSCO-trained Augustus models (see Waterhouse et al., 2018), as well as helixer (Stiehler et al., 2020), GeneMark-ES (Lomsadze et al., 2005), and SNAP (Korf, 2004) outputs. We used tblastn to confirm that duplicates in this species were encoded at different loci.

BUSCO analysis

Each species' genome was queried for the 954 genes of the metazoan BUSCO v.5 gene set (Manni et al., 2021; Simão et al., 2015) using the Saga supercomputer cluster managed by Sigma2 and the Norwegian Research Infrastructure Services. To avoid redundancy, only the 909 BUSCO genes that do not encode regulatory RBPs, TFs, or DNARPs were used to calculate losses (Supplementary File 1, tabs 6 and 7).

Tests for systematic error

We tested for systematic error arising from genome quality issues and mutational bias. To test genome quality, we first regressed BUSCO loss and average regulatory loss against the proportion of false-negative miRNAs (i.e., conserved mature miRNA sequences present in an sRNA dataset without a corresponding genomic sequence) because genomes with more false negatives are of a poorer quality. Second, we regressed BUSCO loss against genome scaffold N50 values because genomes with higher N50 values are composed of longer continuous sequences, and so are higher quality. In these tests, a correlation between loss and quality would indicate that losses were systematically biased by assembly quality. Finally, we regressed average regulatory loss against the branch length leading to each species in the tree generated in our molecular clock analysis (see below). Errors in genome assembly that would create spurious gene losses would not cause substitutions in present gene sequences, so a strong correlation between these variables would imply that losses were not biased by assembly quality.

Natsidis et al. (2021) showed that automatic protein annotation pipelines spuriously reassign ancestral genes to novel orthogroups when mutation rates are high, systematically overestimating both losses and gains. In our dataset, novel miRNA orthogroups were permitted, so extensive mutation would create a correlation between losses and gains. Though only novel paralogues within ancestral TF, RBP, and DNARP orthogroups were permitted, extensive mutational bias would still create a loss–gain correlation. TFs, RBPs, and DNARPs have functionally critical domains that are robustly conserved and often shared among several related orthogroups (Aravind & Koonin, 1998; Aravind et al., 1999; Banjerjee-Basu & Baxevanis, 2001; Bergholtz et al., 2001; Bork et al., 1997; Bustin et al., 1990; Callebaut & Mornon, 1997; Gerstberger et al., 2014; Harrison, 1991; Lowry & Atchley, 2000; Mason et al., 2014; Matia-González et al., 2015; McGinnis et al., 1984; Pabo & Sauer, 1992; Scott & Weiner, 1984; Soulard et al., 1993; Swanson et al., 1987; Wolberger, 1993; Zahler et al., 1992; Zamore et al., 1997). We recorded all significant BLAST hits encoded at distinct loci, so even if extensive mutation obscured a sequence's true orthogroup (thereby creating a spurious loss), it would not be ignored entirely; instead, it would most likely be curated as a paralogue in a related orthogroup. Thus, we still expect a correlation between protein losses and gains if mutation biased our dataset. We tested for these correlations by regressing percent miRNA, TF, RBP, and DNARP losses against percent gains (Supplementary File 1, tab 8). Losses were negative-valued, so a negative regression slope would indicate that species with high loss also had high gain.

Comparison of losses between categories

To examine the relative degree of loss across different categories of genes, we regressed the average of the percent losses of TFs, RBPs, miRNAs, and DNARPs in each species against their percent loss of BUSCO genes. We tested for differences in the slope of this regression according to a species' broad ecology (parasitic or free-living), particular strategy (the six Poulin strategies, or free-living), definitive host (plant, vertebrate, arthropod, or free-living), and parasitic lifecycle stage (larva, adult, lifelong, or free-living). We confirmed the robustness of the relationship between losses across categories using phylogenetic independent contrasts. Losses at ancestral nodes were not linearly interpolated from losses in the descendants thereof (see below), so we did not use Felsenstein's (1985) branch length correction.

Enhanced losses within functions

Following Cunha et al. (2023), we performed Gene Ontology (GO) enrichment analyses on the set of BUSCO genes lost in each of our species. Cellular component GO terms for each gene were obtained from OrthoDB v12.2 (Tegenfeldt et al., 2025) or UniProt r2026_01 (UniProt Consortium, 2025) when the former lacked this information (Supplementary File 1, tab 9). The ancestors of manually extracted terms were obtained using the clusterProfiler (Yu, 2024) and GO.db (Carlson, 2026) packages. Ancestral terms which were overly broad or contained exactly the same genes as one of their descendants were removed to limit the number of nonindependent hypotheses tested. We tested enrichment using clusterProfiler's enricher function with default p- and q-value thresholds and a minGSSize threshold of 5. GO terms are not independent, so clusterProfiler's multiple hypothesis correction may be overconservative. To test the robustness of our results, we approximated the nonindependence of GO terms as the average number of GO terms mapped to each BUSCO gene and re-ran the analyses with p- and q-values equal to 0.05 multiplied by this factor.

Patterns of gene identity

To determine whether ecology influenced the set of genes that were lost in different species, we restricted our dataset to include only those sampled genes present in the last common ancestor of bilaterians and recoded all genes as present or absent in each species.

To examine differences in aggregate gene identity (i.e., the collection of genes retained or lost in each species), pairwise distances between species were calculated from the gene presence matrix using the Jaccard index in the vegan v2.7-1 (vegandevs.github.io/vegan) package. The Jaccard index disregards shared "0" values, so we used a matrix in which "1" indicated the presence of a gene to group species by their retained genes, and a matrix in which "1" indicated the absence of a gene to group species by their lost genes. In the latter matrix, a placeholder gene equal to "1" in the last common ancestor of bilaterians was created so this species was not an empty row. The last common ancestors of spiralians and nematoides were also included in these analyses. Each distance matrix was summarized with a principal coordinate analysis in the ape v5.0 package (Paradis & Schliep, 2019), with a Lingoes correction. Species' positions on the first two principal axes were exported, recentered on the bilaterian ancestor, and plotted in MatLab (Supplementary File 1, tab 10). Statistical support for differences in group distributions was assessed using permutational analysis-of-variance (PERMANOVA) analyses of the Jaccard distances with 100,000 permutations, with loss-only GPC1 values (see below) used to control for total loss in each genome. The "extract.biplot.pcoa" function (gist.github.com/marine-ecologist/9b14015fc8fa0f228c39b8ad0993537c) extracted gene loadings.

To identify individual genes that were disproportionately lost in species with different strategies, we performed Fisher's exact test on contingency tables for each gene counting the number of species of each strategy that had retained or lost it. To account for multiple hypothesis testing, significance was adjusted with a Bonferroni correction (i.e., 0.05 divided by 1337). This correction is overly conservative if gene losses are nonindependent, so a weaker threshold (i.e., 0.05 divided by 133.7) was also used. This procedure was also used to test for nonrandom patterns of loss with respect to a species' parasitic lifecycle stage, its definitive host, and its phylum. The $-\log_{10}$ of the p-value for each gene was visualized with the qqman package (Turner, 2018).

Evolutionary rates

To compare rates of change across taxa, the percent losses of miRNA, RBP, TF, BUSCO, and DNARP genes for all species were summarized using a principal component analysis (PCA). The first principal component (GPC1) accounted for 87.39% of variance and loaded with losses across all categories (Supplementary File 1, tab 11). GPC1 values were mapped onto a time-calibrated phylogeny constructed using the following procedure. First, the predicted amino acid sequences of 19 non-BUSCO housekeeping genes were concatenated and aligned using the Muscle algorithm in MacVector (18.2.5) under the default settings, with minor adjustments made by eye. Branch lengths were calculated individually for all in-group taxa, along with the outgroups *Amphimedon queenslandica*, *Corticium candelabrum*, and *Nematostella vectensis*, using the maximum likelihood criterion in PAUP 4.0a (Swofford, 2002) with a JTT matrix with four rate categories and a Gamma distribution with shape parameter = 0.5. Divergence times were calculated in MEGA7 (Kumar et al., 2016) using the default settings and eight well-constrained calibration points (Erwin et al., 2011) based on an accepted topology of these taxa (Bernot et al., 2023; Laumer et al., 2019; van Megen et al., 2009) built in MacClade v.4.08 (Maddison and Maddison, 2005) (Supplementary File 2, Figure S1).

Gene loss is irreversible on this study's timescales (Marshall et al., 1994), but maximum likelihood methods repeatedly inferred ancestors with more loss than their descendants. To address this error, ancestral gene complements were reconstructed using Dollo parsimony (i.e., all ancestral bilaterian genes present in a tip species must have been present in every ancestral node leading to it). Losses of miRNA families for the last common ancestor of the nematodes *Bursaphelenchus xylophilus* and *Aphelenchus avenae* were calculated by multiplying the number of losses in *B. xylophilus* by the average fraction of non-miRNA regulatory losses shared between these species (89%). Ancestral states were converted into GPC1 values using the PCA factor loadings.

The GPC1 values were mapped onto the time-calibrated phylogeny using the phytools 2.0 package (Revell, 2024), with ancestral species added using the "bind.tip" function with a branch length of 0.00001 Ma. Models of rate change were created that assigned different rates of loss to parasites of different strategies and to all parasites and all free-living organisms. We also constructed models in which the rate of loss changed before the advent of parasitism by assigning rate shifts to a few lineages with high degrees of loss and then adding and removing additional shifts until the corrected Akaike information criterion (AICc, where more negative values indicate better fit) could not be significantly increased (i.e., ΔAICc ≤ -2) by changing the rate of loss for any single species or clade. Shifts corresponding to a ΔAICc ≥ -2 were not included in these local optimum models. Model fit was assessed using the phytools and ape packages.

Results

To test for convergent tempos and modes of gene loss among Poulin's parasitic strategies, we added the hand-curated regulatory and DNA-repair complements of 12 parasitic and 10 free-living species to those of the 8 parasitic and 18 free-living invertebrates in the Peterson et al. (2026) dataset. With these additions, the dataset contained more than 32,000 regulatory and DNA-repair sequences from 48 species (Supplementary File 1, tabs 2–5).

Confirming the reality of losses

We first needed to establish that the observed losses were unlikely to be the result of systematic errors. We assessed the influence of genome assembly problems with three metrics. First, we tested for a correlation between miRNA false negatives (i.e., miRNA sequences present in an sRNA dataset that lacked a corresponding genomic sequence) and genome completeness. Of the 3,962 miRNA genes curated from all taxa with sRNA reads, we found only 45 false negatives, and false negatives were more than 5% of total miRNA genes only in the tardigrade *Paramacrobiotus metropolitanus*, the ostracod *Darwinula stevensoni*, the water strider *Gerris buenoi*, and the crab-hacker barnacle *Sacculina carcini* (Supplementary File 1, tab 12). Further, though miRNA family losses were highly correlated with both average regulatory and repair loss ($R^2 = 0.7167$, $F_{46} = 116.4$, $p = 0.000$) and BUSCO loss ($R^2 = 0.7029$, $F_{46} = 108.8$, $p = 0.000$), the fraction of false negatives was not significantly correlated with either measure of loss (average regulatory and repair: $R^2 = 0.0004$, $F_{44} = 0.0174$, $p = 0.8956$; BUSCO: $R^2 = 0.0183$, $F_{44} = 0.8206$, $p = 0.3699$). Second, we compared each genome's scaffold N50 value to its BUSCO loss. We found no significant correlation between BUSCO losses and N50 values ($cor_{46} = 0.2387, p = 0.1022$; Figure 2A; see also Jauhal & Newcomb, 2021, and Simão et al., 2015), and several species with high N50 values also had high BUSCO losses. Finally, we regressed loss against the branch lengths generated in a molecular clock analysis of the concatenated sequences of 19 non-BUSCO housekeeping genes (total length of nearly 11,500 amino acids). We found a strong correlation between the length of the branch leading to a species and that species' average regulatory and repair loss (Figure 2B; Supplementary File 1, tab 13; see also Peterson et al., 2026, and Wyder et al., 2007), which is unlikely to have been caused by assembly errors.

The correlation between gene loss and branch lengths raised the possibility that divergent sequence evolution could obscure gene identity, systematically inflating both gene loss and gain. To examine this potential bias, we regressed gene losses against gene gains. We did not find a statistically significant relationship between losses and gains in our regulatory protein data (Supplementary File 1, tab 14; $p_{TF} = 0.1415$, $p_{RBP} = 0.7059$, $p_{DNARP} = 0.1409$). We found a significant relationship between losses and gains of miRNAs ($p = 0.0299$). However, because all losses are negative-valued, the positive slope of this relationship ($m = 0.0735$) indicated that, on average, species with high rates of gain had low rates

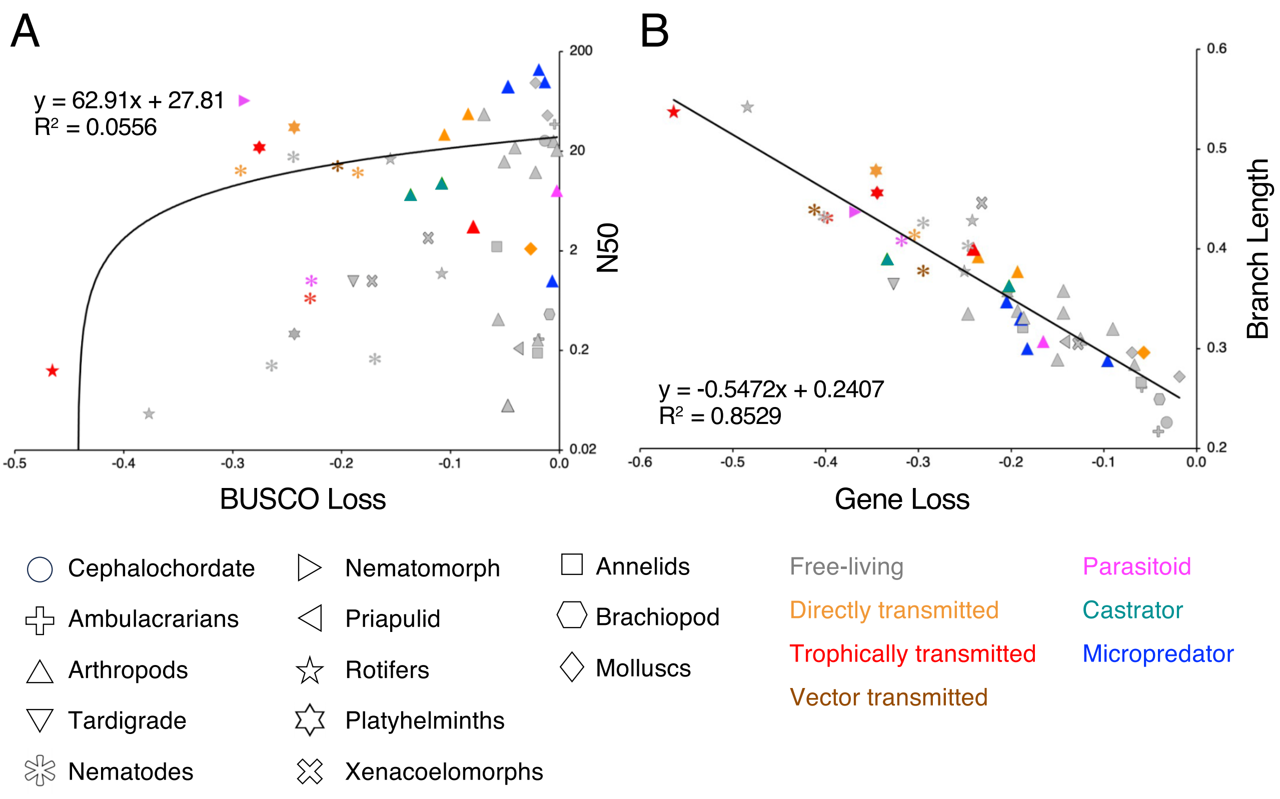

Figure 2. Tests for the influence of assembly errors on loss measurements. (A) The regression of each species' N50 genome value against its BUSCO loss value. All else equal, genomes with higher N50 values are composed of longer continuous stretches of DNA and thus are of a higher quality than genomes with lower N50 values. Loss is not correlated with N50, so the observed losses are unlikely to be the result of errors in genome assembly. A nonlinear fit is used here to account for the logarithmic y-axis. (B) The regression of the branch length leading to each species against the average of regulatory and repair gene loss in that species. That these two values are so strongly correlated suggests that observed gene losses are not primarily the result of random genome assembly errors as these could not produce substitutions in the sequences of present genes.

of loss. When we removed species that experienced whole genome duplications (the horseshoe crab *Limulus polyphemus*, the bdelloid rotifer *Adineta vaga*, and the gooseneck barnacle *Pollicipes pollicipes*) or bursts of miRNA innovation (the octopus *Octopus bimaculoides*), we found no significant relationship between losses and gains for RBPs ($p = 0.3779$), DNARPs ($p = 0.1179$), or miRNAs ($p = 0.0867$). There was a significant relationship between losses and gains of TFs ($p = 0.0233$), but the slope ($m = 0.7554$) again indicated that species with high gains had low losses.

The mode of gene loss

To explore the mode of gene loss in parasites, we first asked whether the ratio of losses between genes with functions that could be supplemented by host genes (i.e., BUSCO housekeeping genes) and genes with functions that could not be supplemented by host genes (i.e., regulatory and DNA-repair genes) differed between parasites with different ecological strategies. Our regression of the average of regulatory and repair loss against BUSCO loss found that these variables were strongly positively correlated (Figure 3). Further, the slope of this regression, which captured the ratio of the losses between these categories, was statistically identical for all parasitic species compared to all free-living ones (Supplementary File 1, tab 15; $F_{44} = 2.031$, $p = 0.1612$), for each of the different parasitic strategies ($F_{34} = 1.195$, $p = 0.3323$), for species that are parasitic at different stages in their lifecycles ($F_{40} = 1.411$, $p = 0.2537$), and for parasites with definitive hosts from different clades ($F_{40} = 0.8765$, $p = 0.4614$). The slope of this relationship among the newly sampled species was not significantly different from the slope among the previously sampled species ($F_{44} = 0.3223$, $p = 0.5731$), suggesting that it is robust to taxon sampling. The correlation of losses across categories remained strong ($R^2 = 0.6688$, $F_{45} = 90.89$, $p = 0.000$) when phylogenetic independent contrasts removed the influence of shared evolutionary history.

To more precisely test for differences in the relative degree of loss among functional categories in parasites with different strategies, we performed GO enrichment analyses of each species' BUSCO losses. In these analyses, 7 of the 20 parasites and 2 of the 28 free-living species displayed losses with significant GO enrichment (Supplementary File 1, tab 16; Figure 4). Thus, BUSCO loss was nonrandom with respect to gene function in only a minority of parasites. Within this minority, enrichment did not converge along ecological lines. For example, the loss of cytoskeletal genes in the nematomorph *Gordionus* sp. (see also Cunha et al., 2023) was paralleled, not in other parasitoids, but in trophically transmitted parasites: the tapeworm *Echinococcus granulosus* and acanthocephalan rotifer *Pomphorhynchus laevis*. Furthermore, no group of terms was frequently enriched across parasites to the exclusion of free-living organisms. Thus, though losses of peroxisome-associated genes occurred in 4 of the 7 parasites (*E. granulosus*, the tongue worm *Rail-*

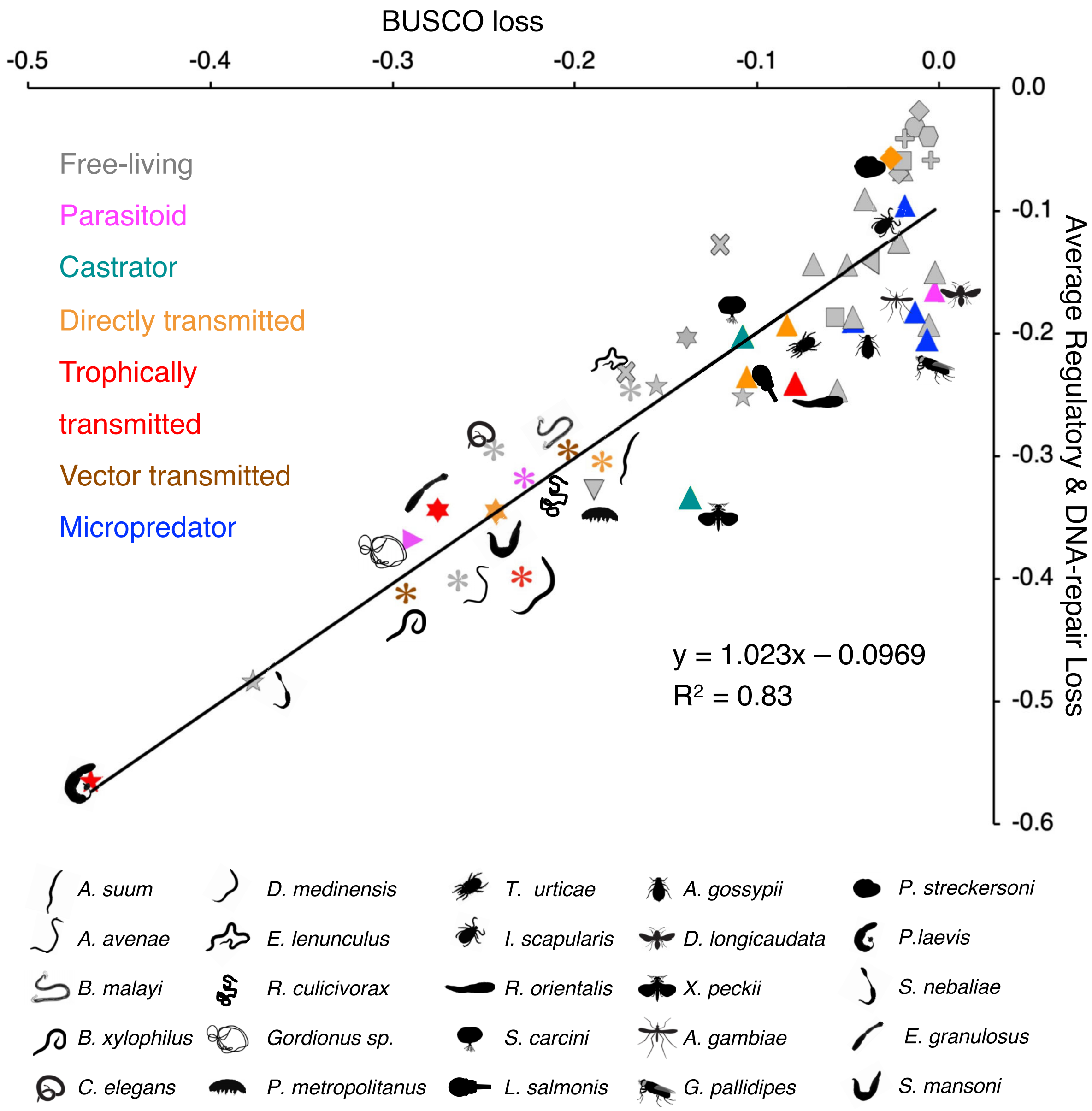

Figure 3. A regression of BUSCO gene loss against the average of regulatory and DNA-repair gene loss for 48 metazoan species. Parasitic and key free-living species are indicated with their respective vignette (bottom). Colors represent the mode of parasitism, with free-living species indicated in gray. There is no significant difference in the slope of the relationship between BUSCO gene loss and average regulatory and repair gene loss for parasitic species, compared to free-living ones. Thus, the basic pattern of gene loss (i.e., the distribution of loss between different functional gene categories) does not differ between parasitic and nonparasitic organisms. Data point shapes and colors as in Figure 2.

lietiella orientalis, the salmon louse *Lepeophtheirus salmonis*, and the blood fluke *Schistosoma mansoni*), they were also seen in the tardigrade *Paramacrobiotus metropolitanus*. Conversely, the loss of endomembrane system genes in *P. laevis* was unique among both free-living and parasitic species. When we relaxed the multiple hypothesis testing correction, significant enrichments were recovered for 10 of the 20 parasites and 9 of the 28 free-living species (Supplementary File 2, Figure S2). These increases notwithstanding, the results were qualitatively similar; enrichments did not converge following the parasitic strategies, and there were not groups of terms that were widely and uniquely enriched among parasites.

Next, we used the presence–absence data for the 1,607 curated ancestral bilaterian genes to create two dissimilarity matrices that acted as aggregate measures of gene identity (i.e., species with similar collections of genes have lower dissimilarities) and summarized these matrices with principal coordinate analyses. For the first matrix, dissimilarities were calculated on the basis of gene presences. The first two principal coordinate axes represented 45.58% of the differences between species in this analysis (Supplementary File 1, tab 10). Among extant organisms, the extremes on the first axis were set by the acorn worm *Saccoglossus kowalevskii* and the acanthocephalan rotifer *Pomphorhynchus laevis*; the extremes on the second axis were set by *P. laevis* and the pine wilt nematode *Bursaphelenchus xylophilus* (Figure 5A; see Supplementary File 2 for discussion of the gene loadings). In the second analysis, dissimilarities were calculated on the basis of gene absences. The first two axes represented

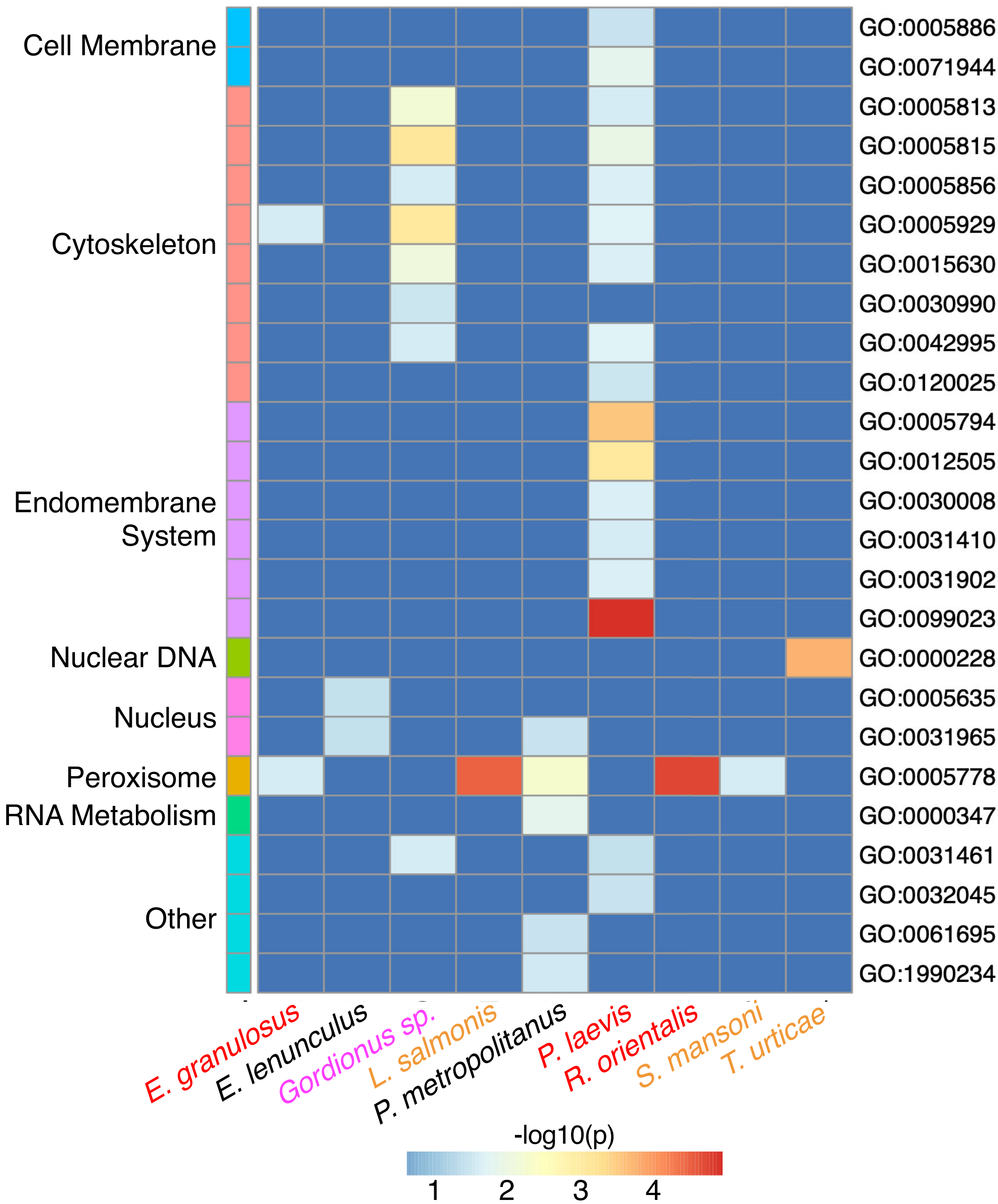

Figure 4. A heatmap of the Gene Ontology (GO) enrichment analysis, under the stringent correction for multiple hypothesis testing, with cells colored according to the −log10 of the *p*-value of the enrichment (higher -log10 values correspond to smaller p-values and, therefore, stronger significance). GO terms that are connected to the same organelle or other cellular structure are grouped together at right; the "Other" category contains transferase and ubiquitinating protein complexes. Though some parasites display losses that are concentrated within particular groups of terms, no groups are consistently and uniquely lost in parasites generally, nor in parasites with a particular ecological strategy. Further, only a minority of parasites display significant enrichments.

28.24% of the differences between species. The extremes on the first axis were set by the brachiopod *Lingula anatina* and *B. xylophilus*; the extremes on the second axis were set by the abalone *Haliotis rufescens* and the fruit fly *Drosophila melanogaster* (Figure 5B). The extreme positions of the fruit fly and other insects reflected the disproportionate loss of DNA-repair genes in their last common ancestor (see also Adams et al., 2000; Mota et al., 2019; Sekelsky, 2017; Wyder et al., 2007).

In the gene-presence analysis, neither parasites collectively, nor the different parasitic strategies separately, occupied a characteristic region of gene-presence space. For example, the increase in the parasitoid nematomorph's Axis 1 score that occurred after its divergence from the nematoid ancestor was associated with a more negative Axis 2 score. This trajectory resembled that of the rotifers and platyhelminths after the origin of the spiralians, despite the nematomorph's ecological similarities to, and close phyloge-

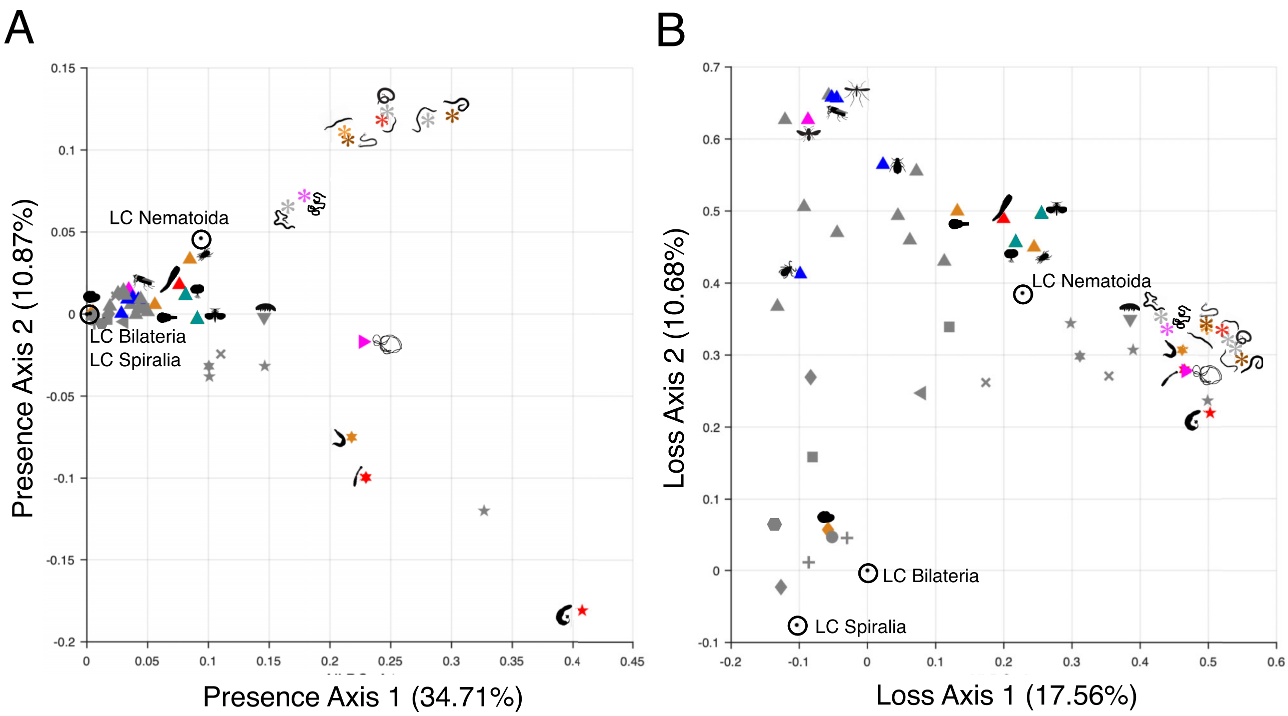

Figure 5. The 48 species in principal coordinate space. (A) The gene-presence space, in which the distances between the points representing each species reflect differences in the subset of the ancestral bilaterian genome which they have retained. The points representing the bilaterian, spiralian, and nematoid last common ancestors are circled and labeled. No clear ecological pattern can be discerned; free-living species are spread out throughout the space, and parasitic species do not appear to cluster according to their strategy. (B) The gene-loss space, in which the distances between the points representing each species reflect differences in the subset of ancestral bilaterian genes which they have lost. Again, no ecological pattern is clearly discernible although many of the species that had the highest total loss, irrespective of whether they are free-living or parasitic, appear to have convergently lost very similar sets of genes. Species silhouettes and datapoint shapes and colors as in Figures 2 and 3.

netic affinities with, the parasitoid nematode *Romanomermis culicivorax*. In turn, this pattern produced a genome that markedly differed from the other two parasitoids. PERMANOVA analyses confirmed that the gene complements of parasites were not significantly more similar to each other than expected by chance, controlling for differences in the total amount of loss between species (Table 2). When the parasites were grouped according to their ecological strategies, similarities between their gene complements remained insignificant.

In the gene-absence analysis, there were again no immediately apparent differences between parasites and free-living species, nor between parasitic strategies. Controlling for the total amount of loss, PERMANOVA analyses found that the sets of genes lost by the parasites were not significantly more similar to each other, or different from free-living species, than expected by chance (Table 2). When the parasites were categorized by strategy, the micropredators were significantly similar to each other. However, when a binary variable equal to "1" for the insects, which accounted for most of the micropredators, was introduced to diminish the phylogenetic bias imposed by their disproportionate loss of DNA repair genes, this similarity became insignificant.

Finally, we examined convergence in the losses of individual genes by performing Fisher's exact test on the number of species in each ecological group which have lost or retained each of the ancestral bilaterian genes. Under the strongest multiple hypothesis testing correction, only the iron-responsive RBP ACO1/IREB2, the ribonuclease ENDOV, and the base-excision repair polymerase POLB displayed a significant relationship between loss and ecology (Figure 6). ENDOV and POLB were disproportionately, though not universally or exclusively, lost in parasites, regardless of strategy, while ACO1/IREB2 was consistently retained by free-living species, parasitoids, and micropredators; consistently lost by trophically transmitted species; and sporadically lost by castrators, directly transmitted species, and vector-transmitted species (Supplementary File 1, tab 17). Under a less stringent correction, five additional proteins displayed significant patterns: the double-strand break repair protein PRKDC, the lysosome-transporting protein BORCS5, the scaffolding protein COMMD2, the ubiquitinating protein LRR1, and the asparagine deamidase NTAN1. Each of these genes was present in most or all micropredators and free-living species; lost in most or all castrators, trophically transmitted species, and vector-transmitted species; and sporadically lost in parasitoids and directly transmitted species. Thus, just 8 of the 1,337 ancestral genes lost in at least one species displayed a pattern of loss significantly associated with ecological strategy, and none were exclusively and uniformly lost by parasites with any particular strategy.

When species were categorized according to their parasitic lifecycle stage, 10 genes displayed significant patterns of losses. ENDOV and POLB were again disproportionately, though not universally or exclusively, lost by parasites, regardless of stage. ACO1/IREB2, PRKDC, LRR1, and BORCS5, along with the zinc-finger TF NXFL1, the endoplasmic reticulum transmembrane protein PARP16, and the

Table 2. Results of the PERMANOVA analysis of gene presence and absence data for all 48 species considered here.

Data set	Comparison	Variable	Degrees of freedom	Sum of squares	R^2	F	p-value
Gene presence	All parasites vs. free-living	Parasite	1	0.0310	0.0232	1.747	0.0889
		Overall loss	1	0.3992	0.2993	22.53	0.00001***
		Residual	48	0.8505	0.6377	—	—
	Parasites by strategy vs. free-living	Parasitoid	1	0.0290	0.0217	1.725	0.1129
		Castrator	1	0.0329	0.0247	1.958	0.1059
		Direct	1	0.0167	0.0126	0.9964	0.3497
		Trophic	1	0.0318	0.0238	1.891	0.0845
		Vector	1	0.0356	0.0267	2.116	0.0877
		Micropredator	1	0.0161	0.0120	0.9555	0.3624
		Overall loss	1	0.3010	0.2256	17.911	0.00001***
		Residual	43	0.7225	0.5417	—	—
Gene absence	All parasites vs. free-living	Parasite	1	0.3047	0.0190	1.089	0.2784
		Overall loss	1	2.046	0.1273	7.308	0.00001***
		Residual	48	13.44	0.8361	—	—
	Parasites by strategy vs. free-living	Parasitoid	1	0.1625	0.0101	0.5714	0.9814
		Castrator	1	0.2871	0.0179	1.009	0.4240
		Direct	1	0.2282	0.0142	0.8023	0.7501
		Trophic	1	0.1810	0.0113	0.6362	0.9555
		Vector	1	0.1533	0.0095	0.5388	0.9741
		Micropredator	1	0.5101	0.0317	1.793	0.0218*
		Overall loss	1	1.646	0.1024	5.787	0.00001***
		Residual	43	12.23	0.7612	—	—
	Unbiased strategies	Micropredator	1	0.157	0.0098	0.5759	0.9771
		Insect	1	0.7847	0.0488	2.879	0.00018**
		Residual	42	11.46	0.7143	—	—

Note. The sum of squares is the amount of variation explained by each factor (residual indicates the unexplained variation); the partial R^2 is the fraction of total variation explained by that factor. The pseudo-F is the ratio of between-group variation to within-group variation; the p-value is the proportion of permuted datasets that yielded a pseudo-F statistic at least as large as that actually observed. For gene absences, the "Unbiased strategies" comparison included effects for the strategies other than micropredation and overall loss, but they did not differ substantially from the results of the "Parasites by strategy vs. free-living" comparison and so are not shown.
*Significant at the 5% level.
**Significant at the 0.05% level.
***Significant at the 0.005% level.

ubiquitinating protein LZTR1 were disproportionately lost by species that were parasitic throughout their lifecycle, relative to free-living species and species that were only parasitic as adults or larvae. Unusually, Mir33 was chiefly lost by permanently parasitic and free-living species but rarely lost among transiently parasitic species. When parasites were categorized according to the clade of their host, 4 genes displayed significant patterns of loss. ENDOV and POLB were disproportionately lost by all parasites, regardless of host clade; ACO1/IREB2 was chiefly lost by parasites of vertebrates; NFXL1 was chiefly lost by parasites of vertebrates and plants. Overall, only 12 of the 1,337 genes analyzed here displayed patterns of loss that were significantly related to at least one major aspect of the ecology of the sampled parasites. In contrast, 228 genes displayed patterns of loss that were significantly related to a species' phylum, regardless of its ecology.

Altogether, these analyses found that the mode of gene loss in parasites, whether it was examined at the level of functional groups, GO categories, total gene complements, or individual genes, was not systematically different from that of free-living organisms.

The tempo of gene loss

To produce a synthetic measure of total loss accounting for the relationship of losses across functional categories, the percent losses in each category were summarized with a PCA (Supplementary File 1, tabs 8 and 11; Supplementary File 2, Figure S3). The first principal component (GPC1) explained 87.19% of the variance in the data and loaded with the loss of genes in all categories. These values, along with reconstructed ancestral states, were then mapped onto a time-calibrated phylogeny of the sampled species (Supplementary File 2, Figure S1).

We first assessed support for a model in which each parasitic strategy had a distinct rate of gene loss. This model assigned rate shifts to the terminal branches in the phylogeny leading to parasitic species, along with the directly transmitted ancestors of the neodermatan platyhelminths and spirurian nematodes (Brabec et al., 2023; van Megen et al., 2009). Because this model required a large number of parameters but fit the data only slightly better than a single-rate Brownian motion model, it was strongly disfavored compared to the single-rate model (Table 3). A model in which all parasites had the same elevated rate of loss performed better than the strategy-based model (ΔAICc = −18.67; relative likelihood = 1) but was still disfavored to the single-rate model.

We also tested a series of models corresponding to the hypothesis that parasites with conspicuously elevated rates of loss simply belonged to fast-evolving clades. We recovered two local optimum models (see the *Materials and Methods* section). The minimal model saw increases in the rate

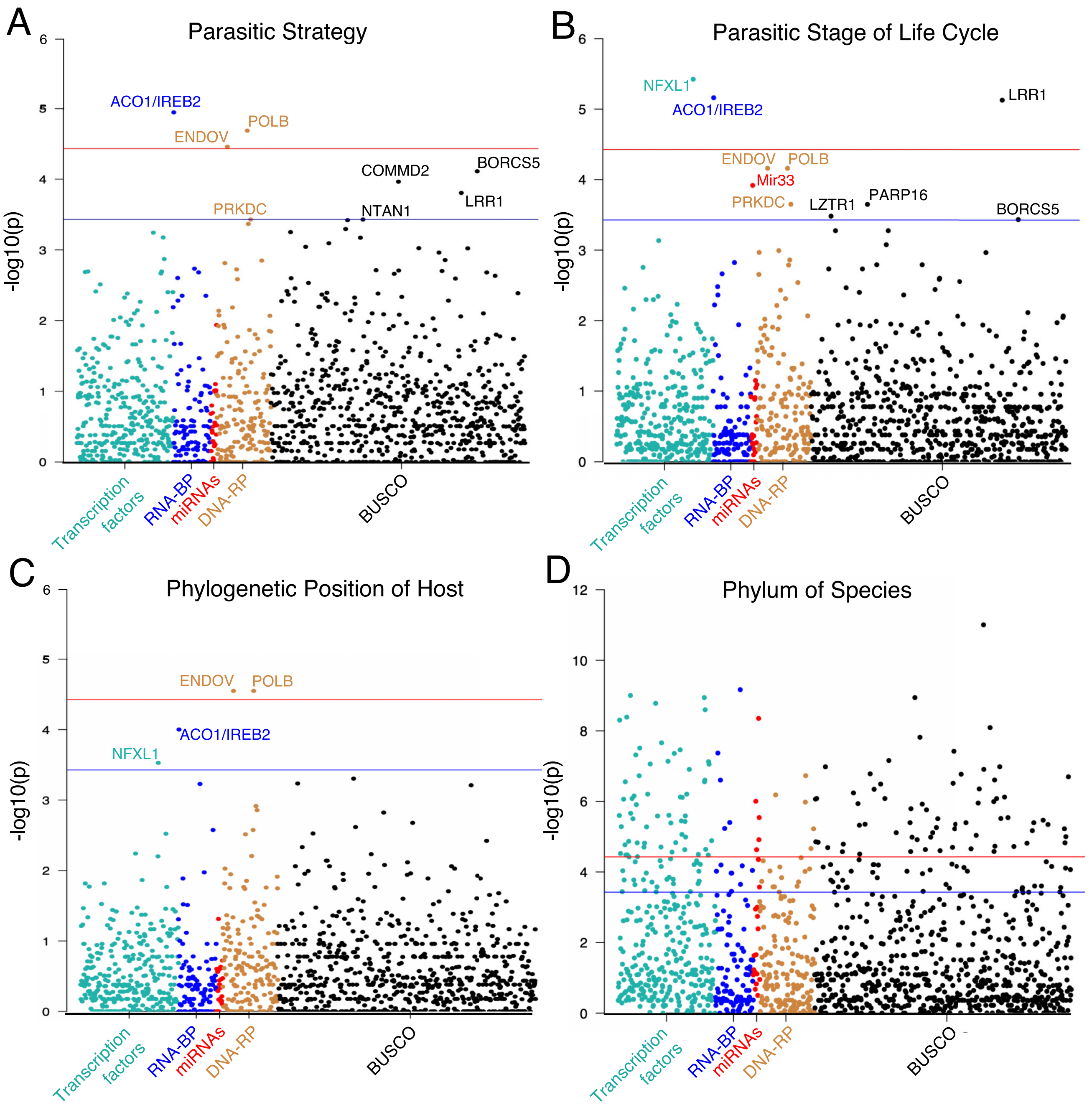

Figure 6. Manhattan plots of the significance of differences in the frequency of gene losses between (A) species with different parasitic strategies, (B) with different parasitic lifecycle stages, and (C) with hosts of different clades. Genes above the red (upper) horizontal line were significant under the strictest correction for multiple hypothesis testing while genes above the blue (lower) horizontal line were significant only under the more relaxed criterion (see the *Materials and Methods* section). Just 10 of the 1,346 ancestral bilaterian genes considered here display patterns of gene loss that are significantly nonrandom with respect to the ecological categories examined in at least one analysis. (D) More than 200 genes display a significantly nonrandom pattern of loss when all species, free-living and parasitic, are categorized according to their phylum.

of gene loss in the stem lineages of all rotifers (labeled "R" in Figure 7), platyhelminths (labeled "P"), and nematoides (labeled "N"), along with the single branches leading to the parasitic strepsipteran *Xenos peckii* and the free-living tardigrade *Paramacrobiotus metropolitanus* (Table 4). The global optimum model (Figure 7) saw additional accelerations in the stem lineages of all oligostracans (labeled "O"), hemipterans (labeled "H"), and xenacoelomorphs (labeled "X"), as well as the single branches leading to the parasitic mite *Tetranychus urticae* and the free-living orbiniid annelid *Dimorphilus gyrocilatus*. The optimum model was slightly, but not significantly, favored over one in which the minimal model lineages evolved faster than other high-rate lineages. In any case, the only parasites that displayed a significantly accelerated rate of gene loss compared to their free-living relatives were the strepsipteran and the mite (branches labeled "1"). Further, several predominantly or exclusively free-living groups (branches labeled "2") displayed an ac-

Table 3. Descriptive statistics for the ecology-focused models tested.

Description	Rates	Shifts	Parameters	Log likelihood	*p*-value relative to null (χ^2)	AICc	ΔAICc relative to null	Relative likelihood	Weight
Different rates for each strategy	7	20	27	63.3671	0.1342	−50.1671	64.7431	0.0000	0.0000
Acceleration in all parasites	2	17	19	58.4829	0.9103	−68.8325	46.0776	0.0000	0.0000
Single-rate Brownian motion	1	0	1	58.4766	—	−114.9101	0.0000	1.0000	1.0000

Note. AICc = corrected Akaike information criterion. The number of parameters is the sum of the number of different rate categories used and the number of rate changes they experienced. The log-likelihood value describes how well the model matches the data without a penalty for increasing model complexity. The AICc accounts for the number of parameters in the model, with more negative AICc values indicating a better fit of the model to the data. ΔAICc for each model is the difference in AICc values between that model and the model with the most negative AICc value (in this case, the null model in which a single-rate Brownian motion process generated the data); more negative ΔAICc values indicate better-fitting models. The relative likelihood uses the ΔAICc to describe the probability of each model compared to the best model, with 0 indicating no probability and 1 indicating equivalent probability. Model weight is the fraction of a model's relative likelihood compared to the sum of relative likelihoods.

Table 4. Descriptive statistics for the locally optimal phylogeny-focused models tested.

Description	Rates	Shifts	Parameters	Log likelihood	*p*-value relative to null	AICc	ΔAICc relative to best AICc	Relative likelihood	Weight
Fast rate in rotifers, platyhelminths, nematoides, *P. metropolitanus*, and *X. peckii*	2	5	7	82.8780	0.0000	−150.4687	13.0974	0.0014	0.0008
Fast rate in rotifers, platyhelminths, nematoides, oligostracans, hemipterans, xenacoelomorphs, *P. metropolitanus*, *X. peckii*, *T. urticae*, and *D. gyrocilatus*	2	10	12	95.6855	0.0000	−163.5661	0.0000	1.0000	0.5567
Fast rate in rotifers, platyhelminths, nematoides, *P. metropolitanus*, and *X. peckii*. Moderate rate in oligostracans, hemipterans, xenacoelomorphs, *T. urticae*, and *D. gyrocilatus*.	3	10	13	96.8003	0.0000	−163.1068	0.4593	0.7948	0.4425

Note. AICc = corrected Akaike information criterion. The contents of each column are as in Table 3, save that the AICc of the null model is much less negative (i.e., far less supported) than those of any of the models here, and thus, was not used to calculate ΔAICc and relative likelihood.

celerated rate; several parasitic species, including endo- and ecto-parasitic representatives of four strategies (branches labeled "3") retained the ancestral bilaterian rate of gene loss; and all of these models were strongly favored over the ecological models (the least negative ΔAICc = −81.64). Thus, the rate of gene loss in bilaterians generally changed independently of parasitism.

Discussion

The reality of losses

We employed four tests to assess the extent of systematic error in our loss measurements. In these analyses, we found no systematic relationship between three independent indicators of genome assembly quality and loss, nor between an indicator of mutational bias and loss. Thus, though it is impossible to truly prove an absence—and our measurements are upper bounds on loss in each species—it is highly unlikely that these measurements are systematically biased across our dataset. That is, there may be some genes in each species that we failed to recover, but the number of false negatives is likely random with respect to the ecology and phylogenetic position of that species and so, contributes noise, not a signal, to our dataset.

The tempo and mode of gene loss

With 20 independent strategic experiments in parasitism from six phyla, representing all of Poulin's strategies (Figure 1 and Table 1), our dataset enables robust comparisons of the mode and tempo of gene loss across a range of clades and ecologies. We found that Poulin's parasitic strategies do not display convergent patterns of gene loss and

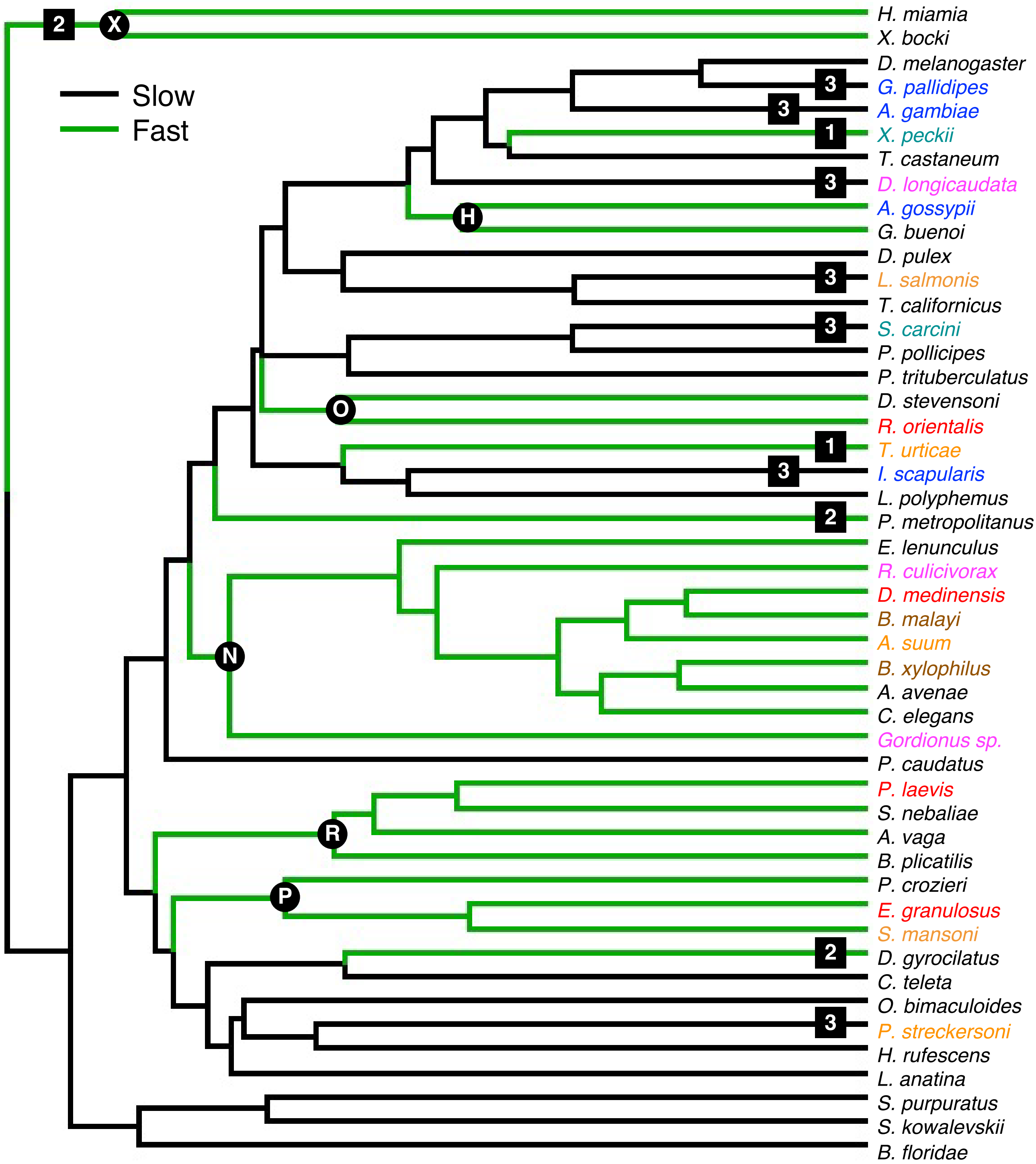

Figure 7. The most strongly supported model of variation in the rate of gene loss across bilaterians. Here, the rate of gene loss independently accelerated in the stem lineages leading to rotifers (R), platyhelminths (P), nematoides (N), oligostracans (O), hemipterans (H), and xenacoelomorphs (X). Only two parasitic species (the strepsipteran *Xenos peckii* and the mite *Tetranychus urticae*, indicated with the "1") display an elevated rate of gene loss that is not shared with their closest sampled free-living relatives. Several lineages with accelerated loss (2) have no or very few known parasitic members, including the tardigrades and orbiniid annelids. Conversely, several parasitic species (3) display the ancestral low rate of gene loss. This model performs only slightly better than one in which the rotifers, platyhelminths, and nematoids, along with the tardigrade and the strepsipteran, evolve even faster than other accelerated lineages.

that parasites as a whole are not defined by a unique pattern of gene loss relative to free-living species. Regarding the mode of loss, parasites of all strategies displayed the same relationship between BUSCO loss and regulatory and repair loss as free-living species and did not share characteristic enrichments of losses among genes associated with particular GO terms. Furthermore, none of the parasitic strategies were characterized by the loss or retention of a particular subset of the ancestral bilaterian genome, and very few of the 1,337 ancestral genes lost in at least one species displayed an ecologically conditioned pattern of loss.

Regarding the rate of loss, almost all parasites with an accelerated rate of gene loss compared to the ancestral bilaterian state have close free-living relatives with that same rate. We did not sample any secondarily free-living species, so this similarity in the mode and tempo of gene loss between parasites and their free-living relatives reflects changes that occurred in their free-living common ancestors. This interpretation is supported by the fact that several free-living species with accelerated rates of gene loss belong to lineages (e.g., tardigrades) with few known parasitic members, and so are almost certainly primitively free-living. Though many of these lineages are miniaturized, the relationship between miniaturization and gene loss is unclear because *Xenoturbella bockii* has accelerated loss but is not miniaturized, while *Daphnia pulex* is miniaturized but does not have accelerated loss. Thus, there is no obvious distinction in BUSCO, regulatory, or DNA repair gene loss between parasitic and free-living species.

The similarity in the mode of loss between parasitic and free-living species is especially striking. Though a plant-parasitizing nematode may need to be more self-sufficient than a crab-parasitizing barnacle, parasites can likely rely on their hosts for some of the general housekeeping functions performed by the products of the BUSCO genes. However, they almost certainly cannot rely on their hosts for the developmental functions performed by regulatory gene products, nor the repair functions performed by repair gene products. Thus, parasites would be expected to have experienced more BUSCO loss per unit of regulatory loss than free-living organisms or to have experienced enriched loss in particular BUSCO functions. Instead, we see that parasites lose BUSCO and regulatory genes in the same relative proportions as free-living organisms and that there is no group of GO terms in which parasites ubiquitously display enriched loss that does not see similarly enriched loss in at least one free-living species.

The only species for which accelerations in the rate of gene loss could not be shown to precede the rise of parasitism are the strepsipteran *Xenos peckii* and the spider mite *Tetranychus urticae*. The strepsipteran is unambiguously an exception because the beetles (represented here by *Tribolium castaneum*) are its closest free-living relatives (Boussau et al., 2014; Misof et al., 2014; Niehuis et al., 2012). However, whether the mite is an exception is less clear. The internal phylogeny of the chelicerates is unstable, and some free-living mites likely had parasitic ancestors (e.g., Radovsky & Krantz, 1998), so we did not sample any free-living species more closely related to *T. urticae* than the horseshoe crab. Thus, the parasitism-specific acceleration in this species may be a more ancient phenomenon. These exceptions notwithstanding, the predominant pattern across bilaterians is that parasites display the same rate of loss as their closest free-living relatives.

The evolution of parasitism

Altogether, these results suggest that gene loss among parasitic bilaterians is an idiosyncratic phenomenon, with a given parasite's evolutionary history outweighing its ecology, despite the fact that parasitic ecologies are highly stereotyped. An almost universal pattern of phenotypic and ecological convergence overlies a pattern of genotypic contingence, even among the genes regulating the development of the convergent phenotypes. Because the tempo and mode of loss do not converge across parasites, distantly related parasites might not be good genomic models for each other, and broader sampling efforts are needed to better understand parasites as a whole. Our data indicate that an accelerated rate of gene loss can arise even at narrow taxonomic levels, so future work to create a complete picture of parasitism will likely need to be very granular.

As gene loss in parasitic species is not reliably different from loss in free-living ones, it is not clear that loss necessarily makes the acquisition of parasitism irreversible. Gene loss in a parasitic dipteran cannot constrain its evolution if free-living dipterans have experienced similar losses. This does not imply that there are no ecological, morphological, or genomic factors opposing reversion to a free-living lifestyle; however, gene loss generally is unlikely to do so. Therefore, analyses which recover free-living species deeply nested within otherwise parasitic clades should be careful when using Dollo's law (i.e., a complex character cannot be regained once lost) to argue for multiple transitions to parasitism, especially if the parasites in question have unspecialized ectoparasitic associations with their hosts. In such cases, the application of Dollo's law should first be validated by showing that each of the parasitic lineages have acquired independent morphological or physiological adaptations for parasitism, which are all absent in the free-living species, or by showing that each parasitic lineage, but not the free-living species, has lost one or more genes that were already known to be essential to a free-living existence. Other studies have proposed violations of Dollo's law among arthropods (Eggleton & Belshaw, 1992; Klimov & O'Connor, 2013; Radovsky & Krantz, 1998; Sharkey, 2007; Walter & Proctor, 1998), molluscs (Wachtler et al., 2001), and nematodes (Dorris et al., 2002), so there may be more secondarily free-living species than once thought.

Limitations of the study

The chief limitation of this study is taxon inclusion. Our sample of 20 independent strategic experiments in 18 parasitic lineages is sufficient to examine first-order patterns of gene loss across bilaterians, but there may be second-order patterns of convergence between, or parallelism within, lineages that cannot be resolved without a more intense sample of the hundreds of known parasitic lineages (Weinstein & Kuris, 2016). Analogously, the genes we sampled represent only a fraction of the genome. Thus, though we did not find convergent losses of regulatory, repair, and BUSCO genes, there may be convergent losses among other gene categories. Finally, this paper did not attempt to identify the mechanisms which control the relative degree of loss across functional categories or the overall rate of loss. Insects, which, uniquely among the sampled bilaterians, have disproportionately lost DNA repair genes may help resolve these questions.

Supplementary material

Supplementary material is available online at *Evolution*.

Data availability

The presence/absence summaries of the curated protein complements, the raw and processed BUSCO outputs, the GO annotations for the BUSCO genes, and the results of all statistical tests are available in Supplementary File 1. All newly curated protein sequences, along with the scripts used to perform all analyses, the alignment nexus file underlying the molecular clock, and the reconstructed ancestral states with phylogenetic independent contrast calculations have been deposited in Dryad (Clarke et al., 2026; doi.org/10.5061/dryad.4j0zpc8t9). A series of .csv files corresponding to the different tabs of Supplementary File 1 have also been included in the Dryad deposit. All newly curated miRNA complements and newly sequenced sRNA datasets have been deposited in MirGeneDB (mirgenedb.org) and will appear on its main page upon the release of its 3.1 update (expected January 2027). Until the release of this update, these complements and sequences will be available upon request to the MirGeneDB web team and from the Dryad deposit.

Author contributions

A.W.C. and K.J.P. conceived the project and made taxon selection decisions. T.M.F., J.O., and H.G. captured or raised specimens. A.W.C., Ł.W., V.M.P., J.V., A.D., R.R.F., P.M., and B.F. collected raw data. A.W.C. and B.F. performed computational data processing. A.W.C., L.B., A.I.B., and K.J.P. curated data manually. A.W.C. and M.A.M. performed statistical analyses. A.W.C. and K.J.P. wrote the paper, with all authors contributing to the interpretation of results and the final draft of the manuscript.

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Conflict of interest

The authors declare no conflict of interest.

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